WORKED SOLUTION

STEP I 2010, Question 7

Pure • Vectors and Intersecting Lines

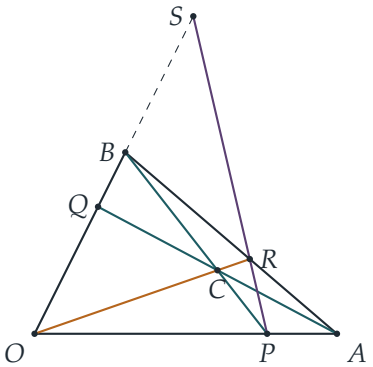
STEP I 2010, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s1-q7

The key idea

Because O, A and B are not collinear, every point of the plane can be written as $\lambda\mathbf{a} + \mu\mathbf{b}$ in exactly one way. So to find where two lines meet, we write a general point of each line in this form and compare coefficients of $\mathbf{a}$ and $\mathbf{b}$. A point lies on the line OB exactly when its $\mathbf{a}$-coefficient is zero, and on AB exactly when its two coefficients add up to 1. Those two facts do almost all the work.



The picture uses $\alpha = \frac{1}{2}$, $\beta = \frac{7}{20}$. Note that S lies on the line OB but outside the segment OB, while Q lies between O and B.

Solution

Finding p and q

A general point of the line BC is

$$\mathbf{b} + \mu(\mathbf{c} - \mathbf{b}) = \mu\alpha\,\mathbf{a} + (1 - \mu(1 - \beta))\mathbf{b}.$$

A point of the line OA has the form $\lambda\mathbf{a}$, with no $\mathbf{b}$ component. Since $\mathbf{a}$ and $\mathbf{b}$ are not parallel (because O, A, B are not collinear), we may compare coefficients. The

$\mathbf{b}$-coefficient must vanish, so $\mu = \dfrac{1}{1-\beta}$; this is allowed because $\beta < 1 - \alpha < 1$, so $1 - \beta \neq 0$. Then

$$\mathbf{p} = \mu\alpha\,\mathbf{a} = \frac{\alpha\mathbf{a}}{1-\beta},$$

as required.

The set-up is symmetric under swapping $A \leftrightarrow B$ and $\alpha \leftrightarrow \beta$ (this swaps $P \leftrightarrow Q$), so we can write down

$$\mathbf{q} = \frac{\beta\mathbf{b}}{1-\alpha}.$$

The point R

First, $\mathbf{r} = \dfrac{1}{\alpha+\beta}\,\mathbf{c}$ is a scalar multiple of $\mathbf{c}$, so R lies on the line OC.

Second, write $\mathbf{r} = t\mathbf{a} + (1-t)\mathbf{b}$ with $t = \dfrac{\alpha}{\alpha+\beta}$. Indeed $1 - t = \dfrac{\beta}{\alpha+\beta}$, which is the coefficient of $\mathbf{b}$ in $\mathbf{r}$. Points of the form $t\mathbf{a} + (1-t)\mathbf{b} = \mathbf{b} + t(\mathbf{a} - \mathbf{b})$ are exactly the points of the line AB, so R lies on AB too.

The point S

A general point of the line PR is $\mathbf{p} + t(\mathbf{r} - \mathbf{p})$, which equals

$$\left[\frac{\alpha}{1-\beta} + t\left(\frac{\alpha}{\alpha+\beta} - \frac{\alpha}{1-\beta}\right)\right]\mathbf{a} \;+\; \frac{t\beta}{\alpha+\beta}\,\mathbf{b}.$$

For this point to lie on OB, the $\mathbf{a}$-coefficient must be zero. Dividing by $\alpha \neq 0$ gives

$$\frac{1}{1-\beta} = t \cdot \frac{(\alpha+\beta) - (1-\beta)}{(1-\beta)(\alpha+\beta)}, \qquad \text{so} \qquad t = \frac{\alpha+\beta}{\alpha+2\beta-1}.$$

(The question tells us the lines meet, which forces $\alpha + 2\beta \neq 1$. If $\alpha + 2\beta = 1$ then PR is parallel to OB.) Hence

$$\mathbf{s} = \frac{t\beta}{\alpha+\beta}\,\mathbf{b} = \frac{\beta}{\alpha+2\beta-1}\,\mathbf{b}.$$

> **Ask yourself** How do we turn a position vector $k\mathbf{b}$ into the ratio of the distances from O and from B?

If X has position vector $k\mathbf{b}$, then $\overrightarrow{OX} = k\mathbf{b}$ and $\overrightarrow{BX} = (k-1)\mathbf{b}$, so

$$\frac{OX}{BX} = \frac{|k|}{|1-k|}.$$

The absolute values matter, because k may be negative or larger than 1, as it is for S in the picture.

The ratios

For Q we have $k = \dfrac{\beta}{1-\alpha}$, and $0 < k < 1$ because $0 < \beta < 1-\alpha$. Then $1-k = \dfrac{1-\alpha-\beta}{1-\alpha}$, so

$$\frac{OQ}{BQ} = \frac{\beta}{1-\alpha-\beta}.$$

For S we have $k = \dfrac{\beta}{\alpha+2\beta-1}$, so

$$1-k = \frac{\alpha+2\beta-1-\beta}{\alpha+2\beta-1} = \frac{\alpha+\beta-1}{\alpha+2\beta-1}, \qquad \frac{k}{1-k} = \frac{\beta}{\alpha+\beta-1} = -\frac{\beta}{1-\alpha-\beta}.$$

Taking absolute values, and using $\beta > 0$ and $1-\alpha-\beta > 0$,

$$\frac{OS}{BS} = \left|\frac{k}{1-k}\right| = \frac{\beta}{1-\alpha-\beta} = \frac{OQ}{BQ}.$$

This completes the proof. The minus sign before we took absolute values says that Q divides OB internally and S divides it externally, in the same ratio.

Checks

- Try $\alpha = \frac{1}{2}$, $\beta = \frac{7}{20}$ (the picture). Then $\mathbf{q} = 0.7\mathbf{b}$ and $\mathbf{s} = 1.75\mathbf{b}$, so $OQ/BQ = 0.7/0.3 = \frac{7}{3}$ and $OS/BS = 1.75/0.75 = \frac{7}{3}$. The formula gives $\beta/(1-\alpha-\beta) = 0.35/0.15 = \frac{7}{3}$.
- For random $\mathbf{a}$, $\mathbf{b}$, α, β, we computed all four intersections directly by solving pairs of line equations and compared them with the formulas, and checked that the two ratios agree. They matched in over 8000 cases.
- Symmetry: swapping $\alpha \leftrightarrow \beta$ and $\mathbf{a} \leftrightarrow \mathbf{b}$ turns the formula for $\mathbf{p}$ into the formula for $\mathbf{q}$, as it should.

Common slips

- Forgetting the absolute values. $\mathbf{s}$ may have a negative coefficient, or one bigger than 1, and a careless $k/(1-k)$ then gives a negative "ratio of lengths".
- Comparing coefficients of $\mathbf{a}$ and $\mathbf{b}$ without saying why it is allowed. It works because $\mathbf{a}$ and $\mathbf{b}$ are not parallel.

Beyond STEP

The points Q and S divide OB internally and externally in the same ratio, so they are *harmonic conjugates* with respect to O and B. The figure here, four lines and their intersections, is the classical *complete quadrilateral*. Harmonic division is preserved by projections, and it is one of the central ideas of projective geometry. The habit of writing

points in terms of two independent vectors and comparing coefficients is the start of linear algebra, in which it becomes the idea of a basis. Strang's *Introduction to Linear Algebra* develops that side.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.